Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 9

Serverless in the Cloud

Microservices, serverless computing and event-driven architectures.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Recap

LAST SESSION

Database

A common cloud service: databases in the cloud.

LAST SESSION

SQL, NoSQL, AWS RDS

Relational and nonrelational databases, and the AWS database services.

Learning objectives

1. Microservices
2. Serverless
3. Event-driven architectures
4. Hybrid methods

THIS WEEK

Serverless

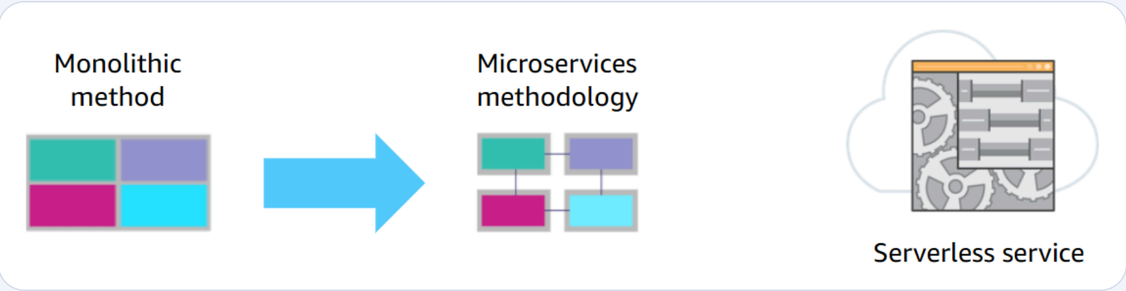
Describe the key serverless concepts.

Introduction to serverless

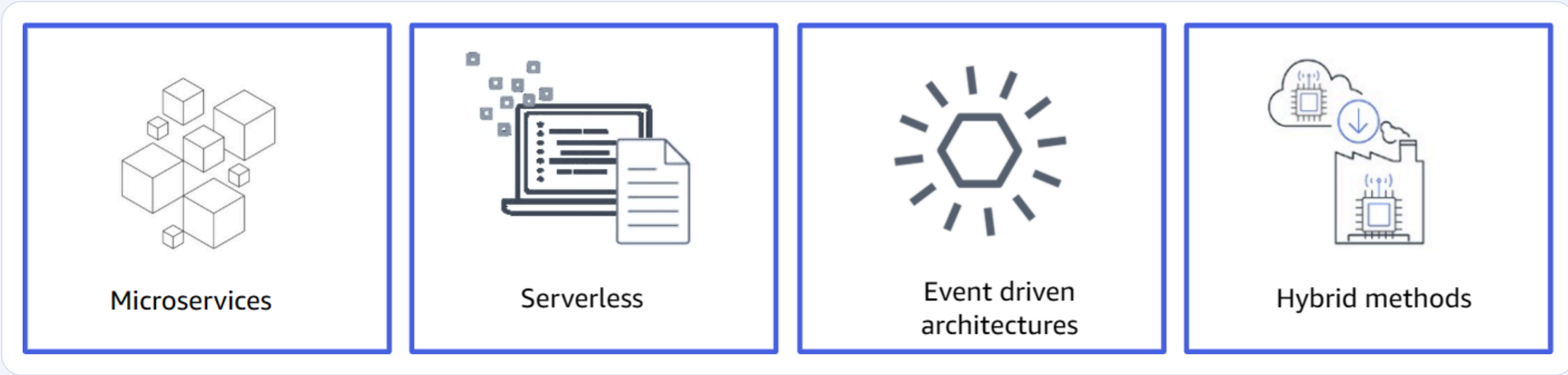
Building a secure, reliable and cost-effective architecture is important for every IT professional.

Many companies no longer build their applications as they did in the past. They are replacing monolithic methods with a microservices methodology.

It is more than a change in how they build. They also have a new development in what they build with: serverless services.



In this section



The basics of microservices, serverless, event-driven architectures and hybrid methods.

PART 1

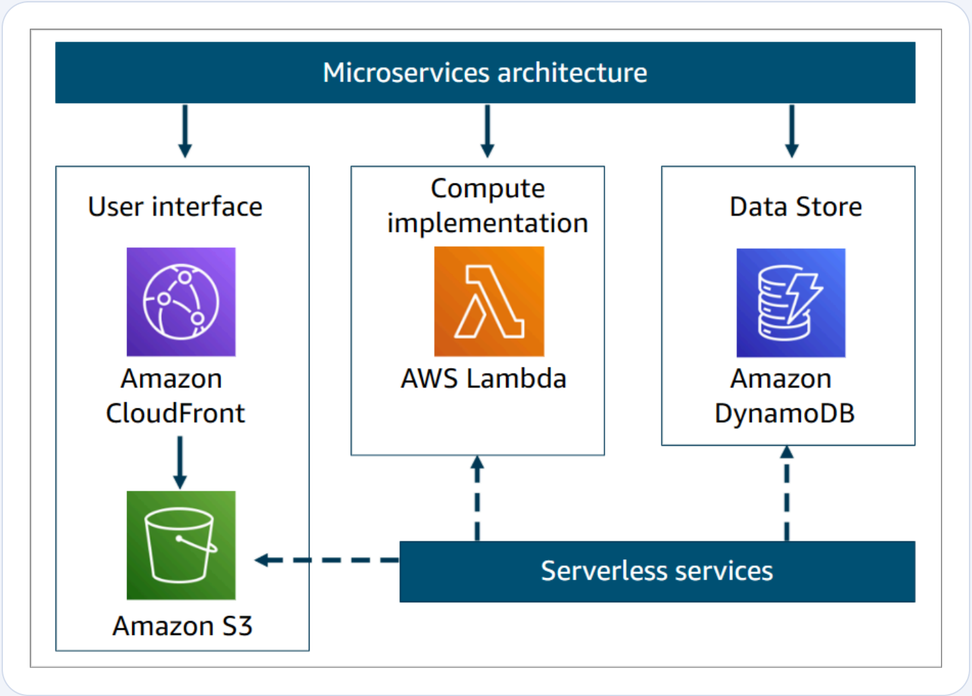
Microservices

Small independent services that communicate over well-defined APIs.

Microservices

Software composed of *small independent services* that communicate over well-defined APIs.

Microservices are an architectural and organizational approach to software development. The services are always owned by small, self-contained teams.

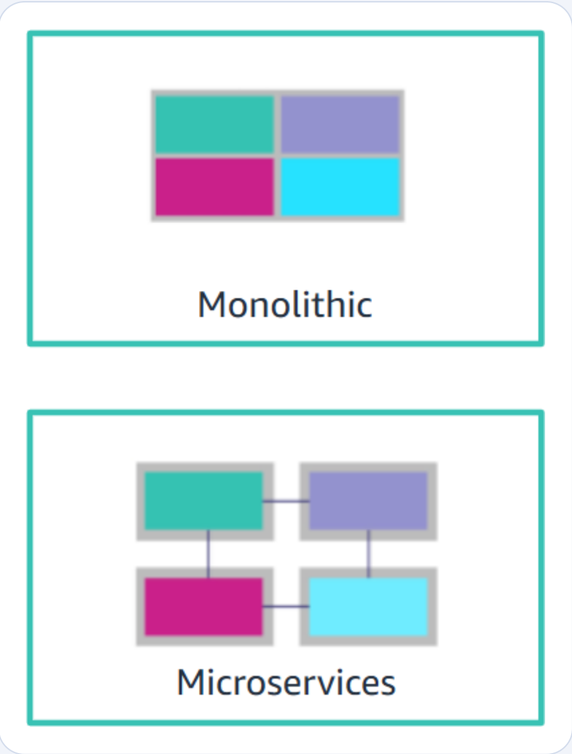


Tightly versus loosely coupled

IT infrastructure can be built with a methodology that is either tightly coupled or loosely coupled.

A tightly coupled architecture is referred to as a monolithic architecture. A loosely coupled architecture is referred to as a microservices architecture.

Microservices is the better approach because your application is more durable, reliable and manageable.

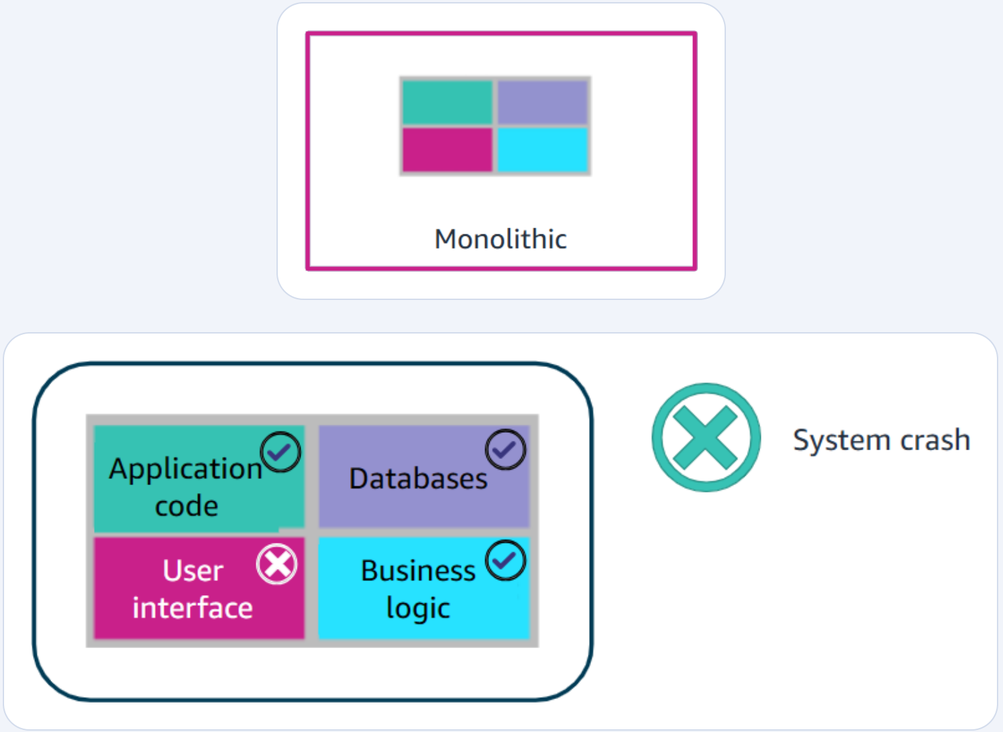


Monolithic application (tightly coupled)

If one piece fails, the entire application crashes.

If there is a spike in demand, the entire architecture must be scaled.

Adding features becomes more complex as time goes on. Pieces of the code base must work with each other to sync properly.

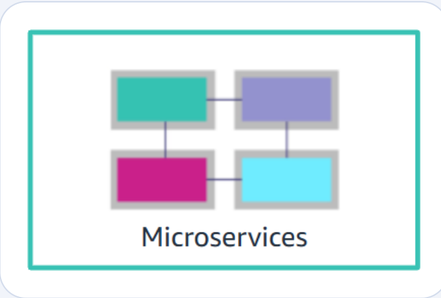


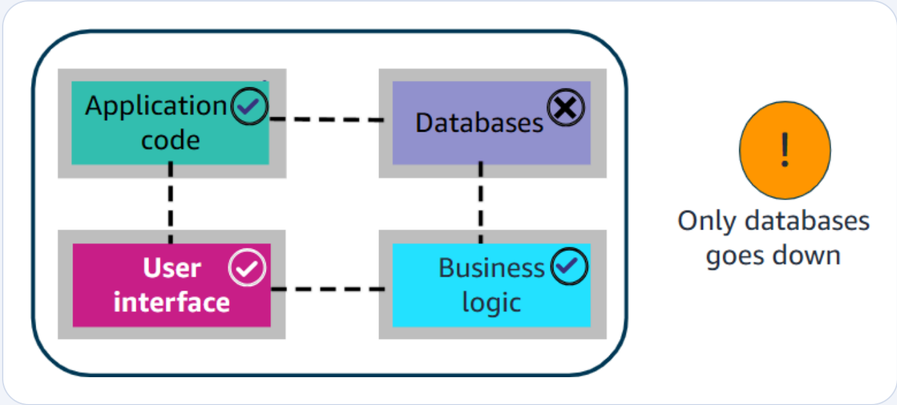
Microservices application (loosely coupled)

An application is built as independent components that run each application process as a service.

Services communicate through a well-defined interface using lightweight APIs. Each is built for a business capability and performs a single function.

Each service can be updated, deployed and scaled on its own. If one component fails, the other components remain active.



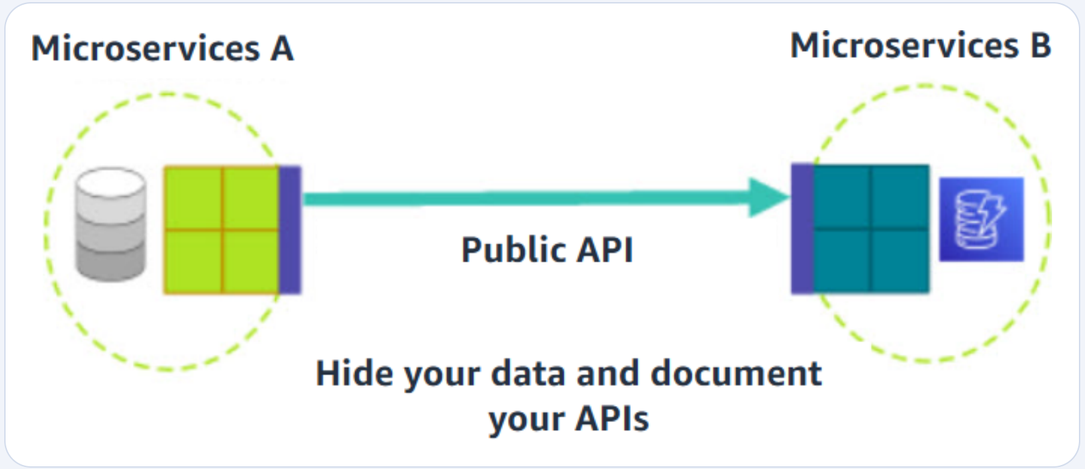


Benefits of microservices: use public APIs

Microservices communicate over APIs. Doing so is the only way to realize the benefits of microservices.

By communicating with APIs, you can modify a database and maintain it without affecting communications or having to repoint interfaces.

The public API is a contract between your microservice and its clients.

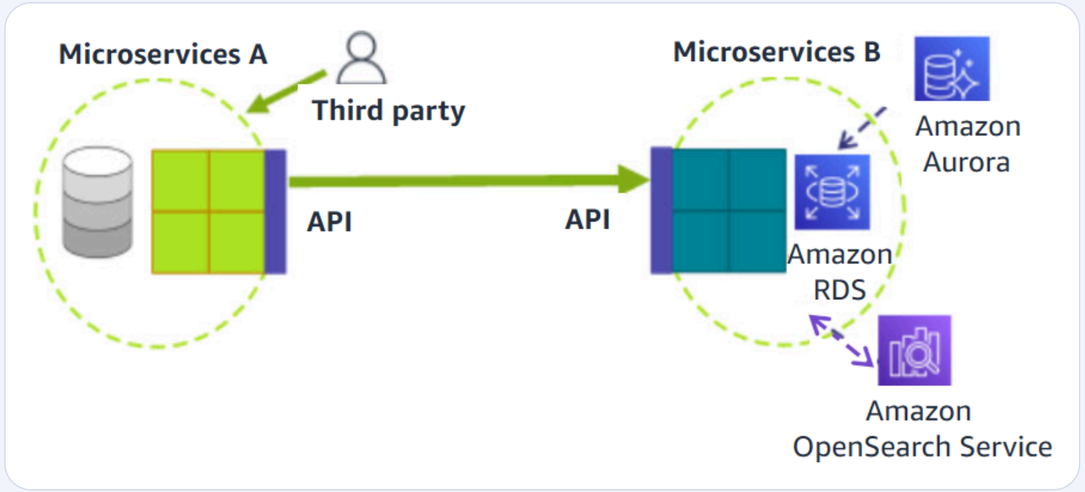


Benefits: the right tool for the job

Because every service is built for explicit functionality, you can choose the right tool for the job.

The team for Microservices B can add Amazon Aurora or change to Amazon OpenSearch Service without affecting Microservices A.

Each team can also code in the language and framework best suited to its service, without depending on other teams.

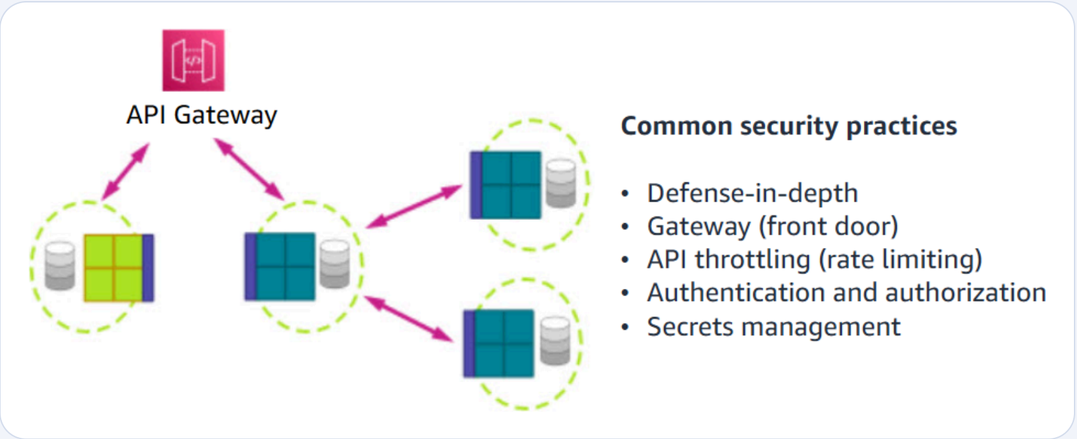


Benefits: secure your services

Microservices must be architected to provide security at every step, an approach called defense in depth.

This covers the application level, the API level, the network layer and AWS Identity and Access Management (IAM) configurations.

The API is the entry point. Amazon API Gateway can authenticate and authorize, enforce throttling and hide internal resources.

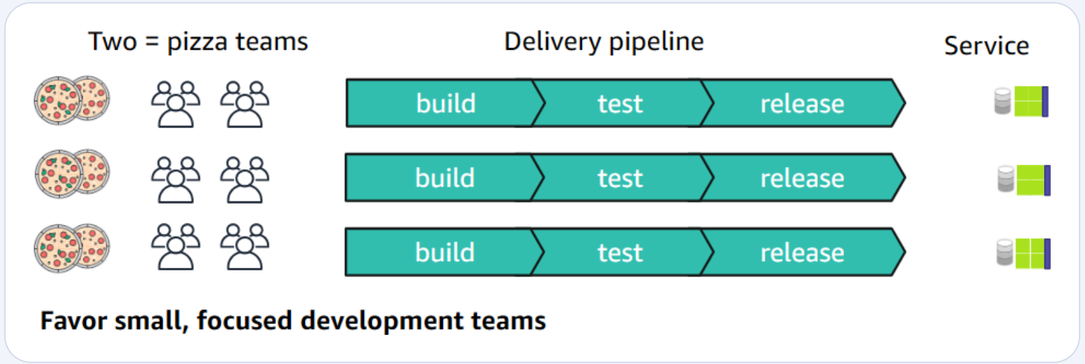


Benefits: company transformation

Going from a monolith to microservices is more than a technological transformation.

Small “two-pizza teams”, no more people than two pizzas can feed, each focus on producing a single microservice.

Each team sends its service through an automated delivery pipeline, so teams are decoupled and release updates on their own.



Downside of monoliths

All processes are tightly coupled and run as a single service.



Adding or improving a monolithic application's features becomes more complex as the code base grows.



Dependent and tightly coupled processes increase the impact of a single process failure.



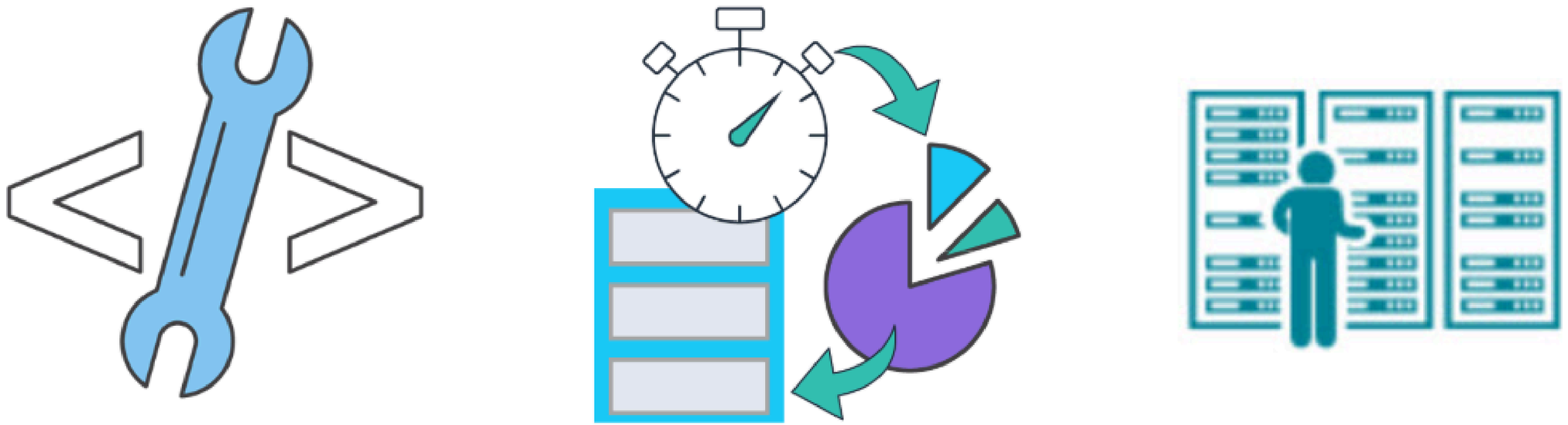
PART 2

Serverless computing

Build and run applications and services without thinking about servers.

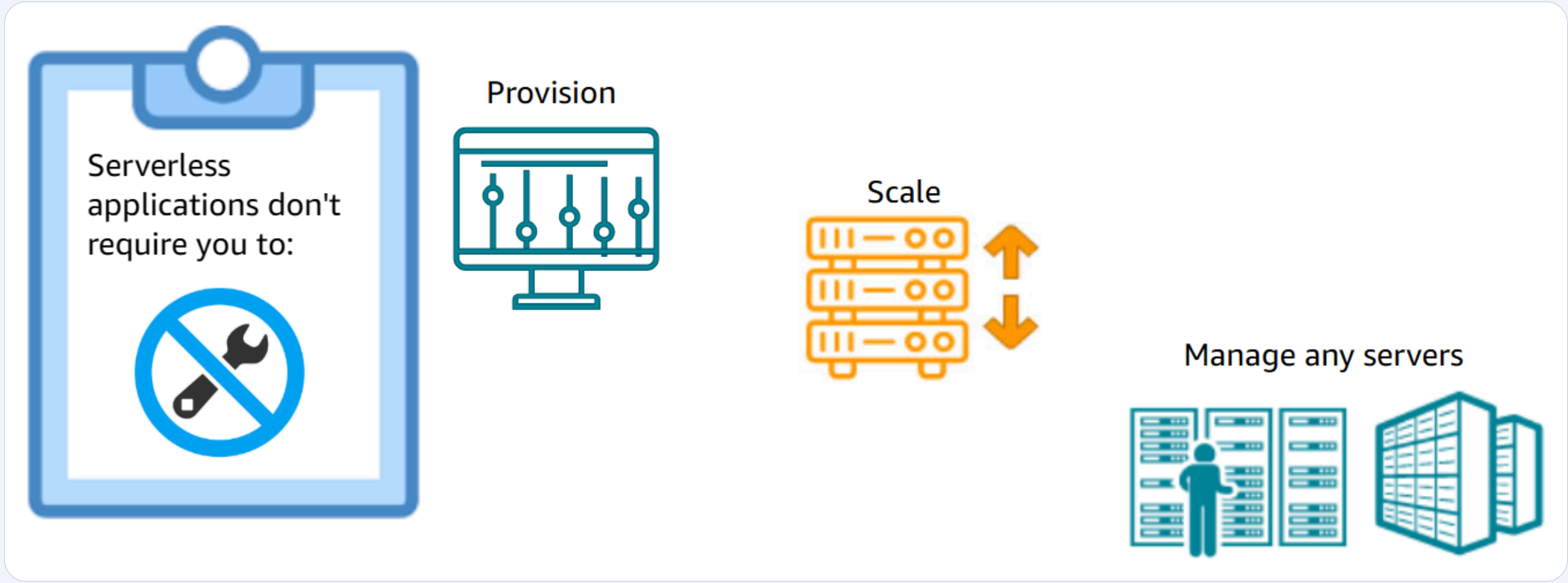
Week 9

What is serverless computing?



Build and run applications and services without thinking about servers.

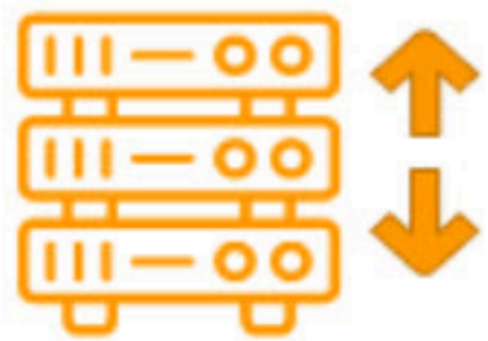
What serverless takes away



Serverless applications do not require you to provision, scale or manage any servers.

What serverless gives you







You can build them for nearly any type of application or backend service, and everything required to run and scale your application with high availability is handled for you.

Why serverless?



Focus on the core product

Developers can focus on their core product.



No servers to run

No worrying about managing and operating servers or runtimes.

Serverless benefits



No server management



Pay for value services



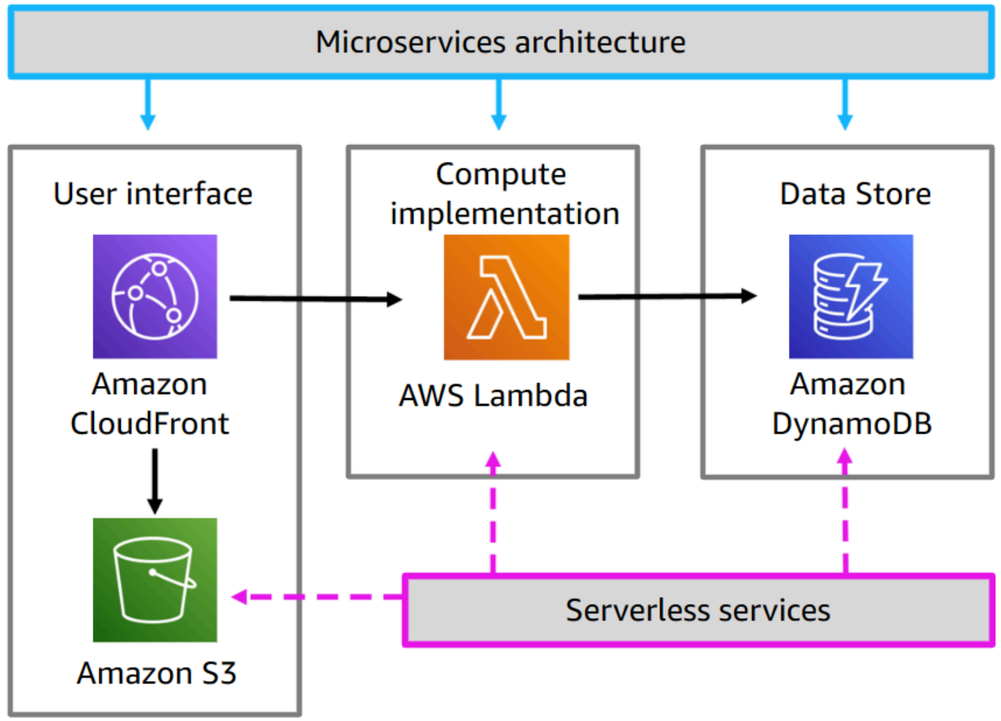
Continuous scaling



High availability

Microservices and serverless architecture

- Microservices is a method of building your architecture.
- Serverless is a service that is used to build your architecture.
- Microservices is the method of loosely coupling sections of your architecture.
- Serverless services are the tools that you use to build out those sections.
- Common to use a blend of both serverless and nonserverless services.

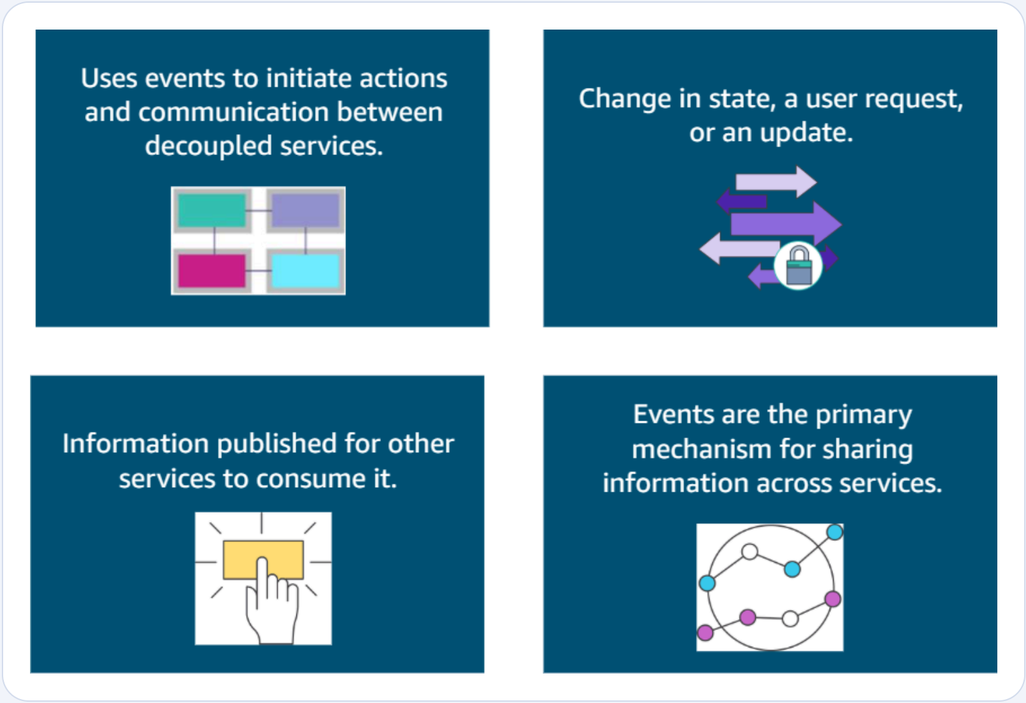


PART 3

Event-driven architectures

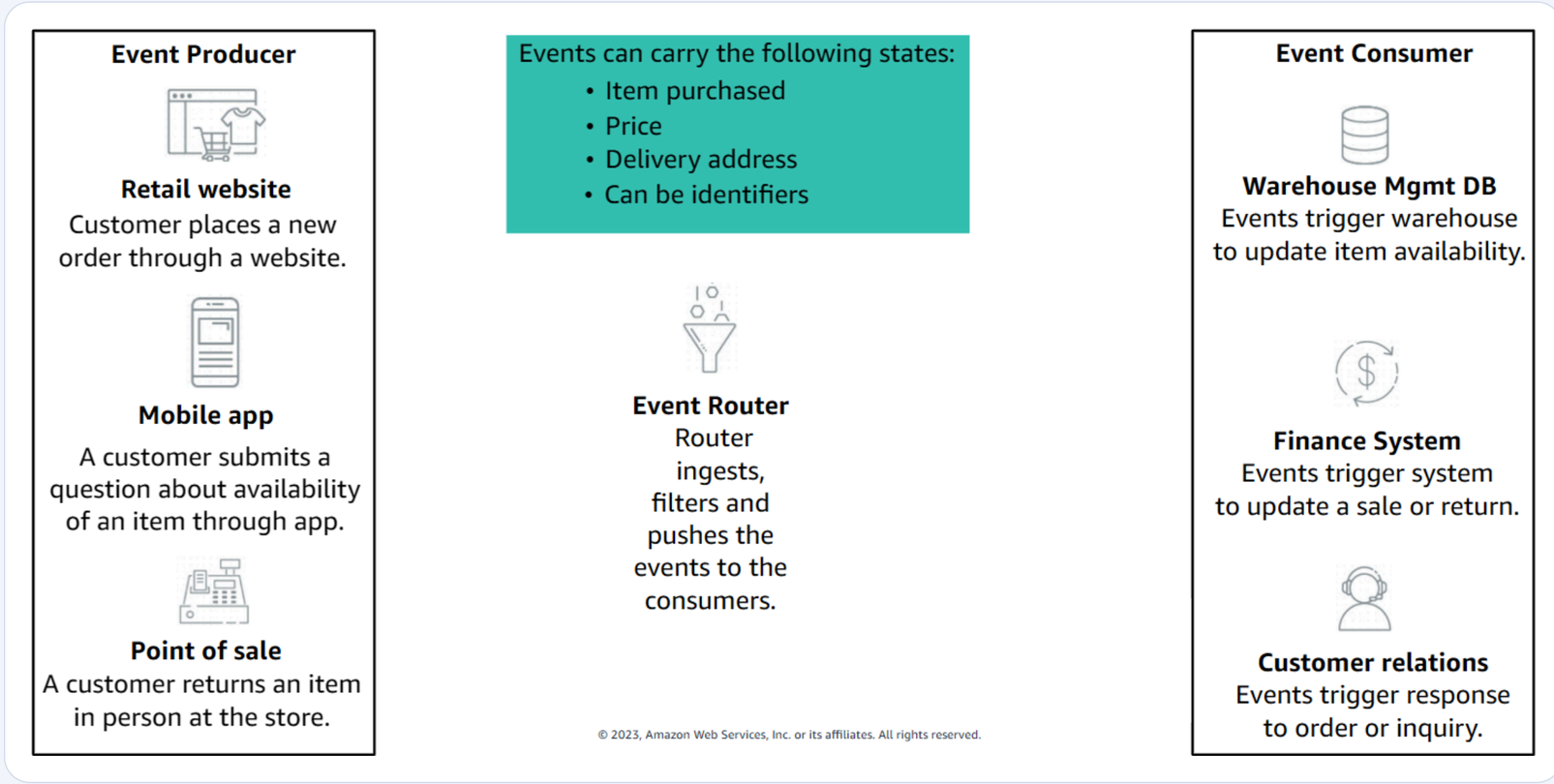
An event is a change in state, a user request or an update.

Event-driven architectures

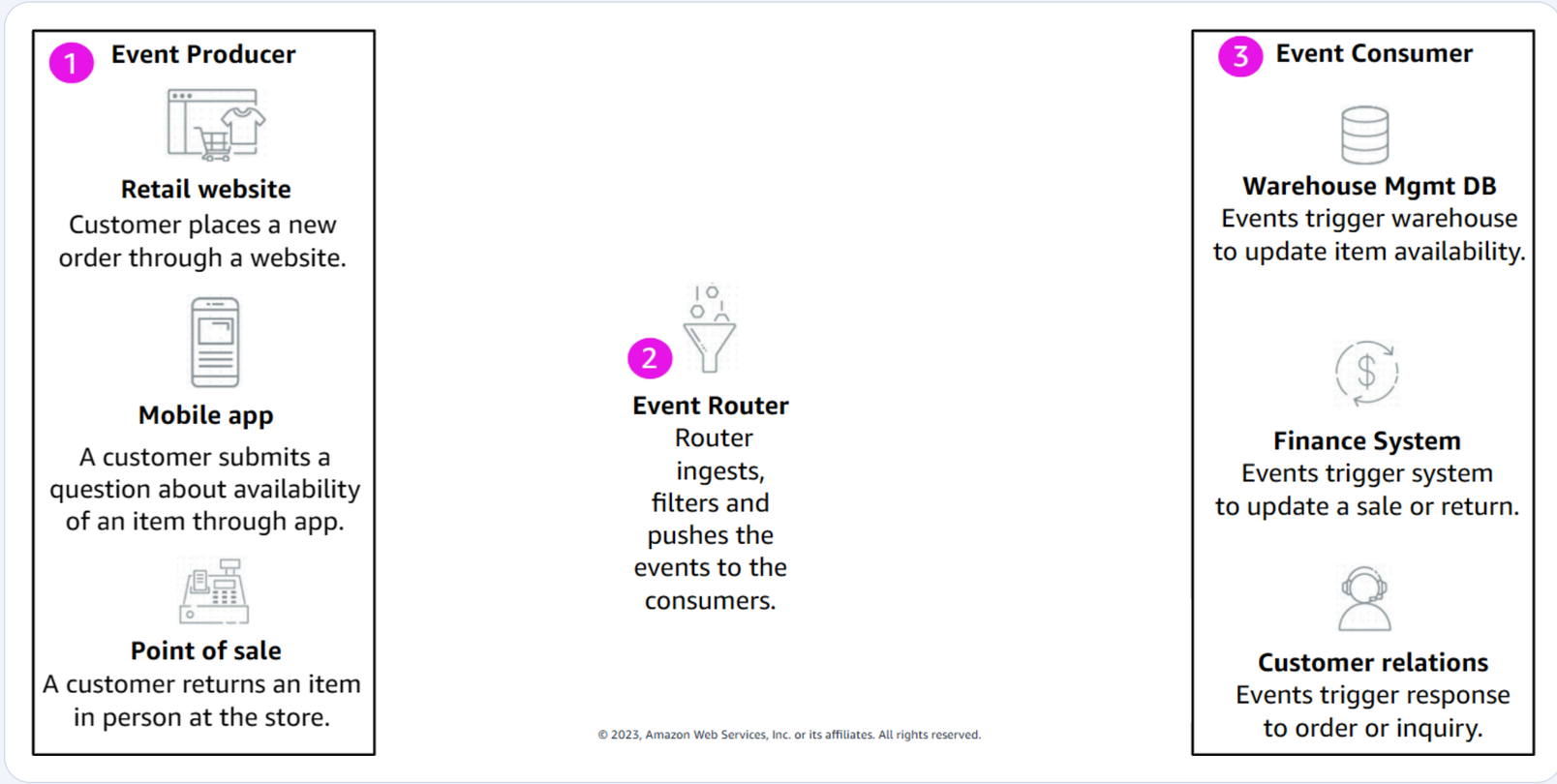


Example of an event: an item being placed in the shopping cart of an e-commerce website.

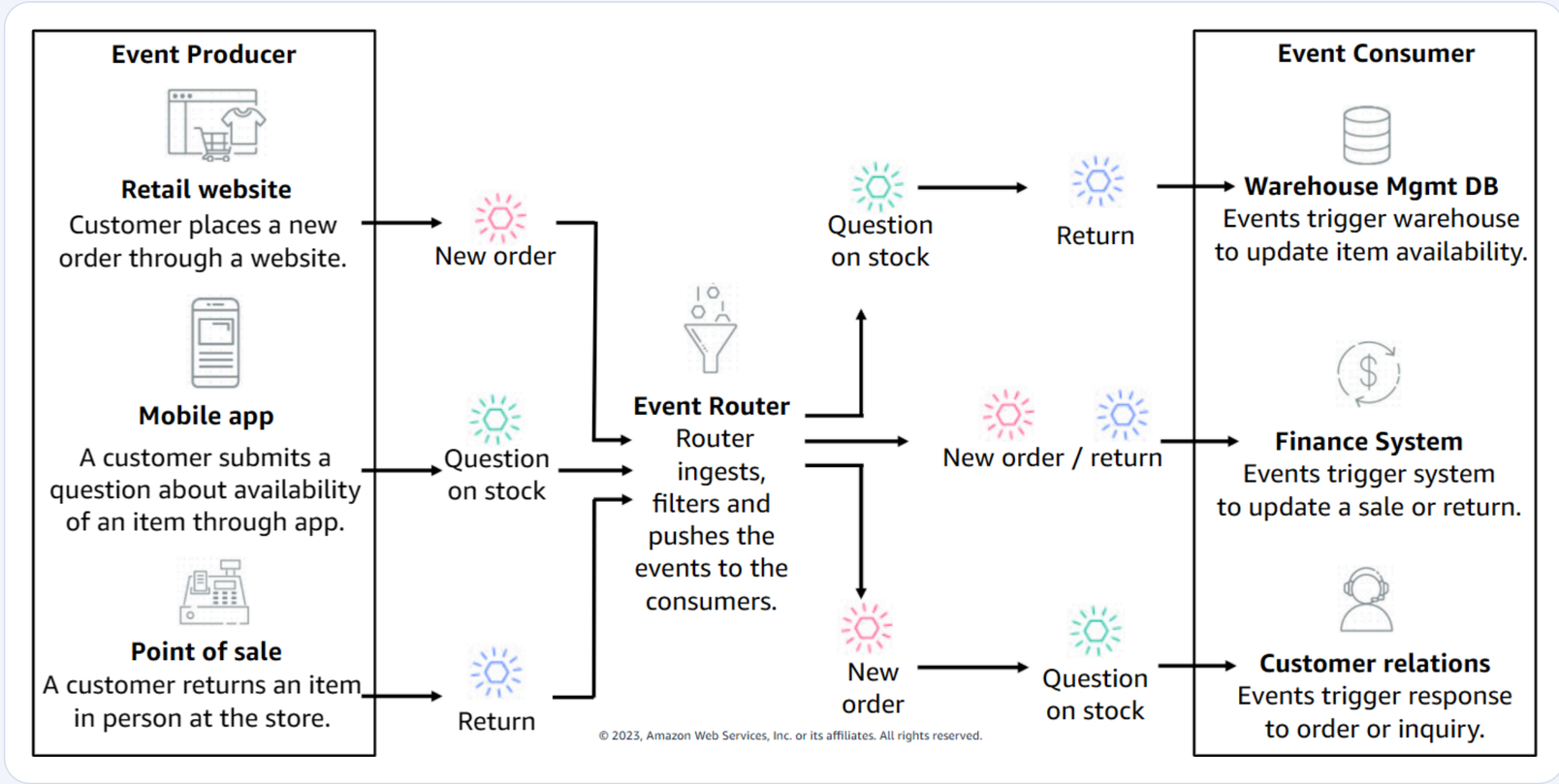
Event-driven architecture diagram: the parts



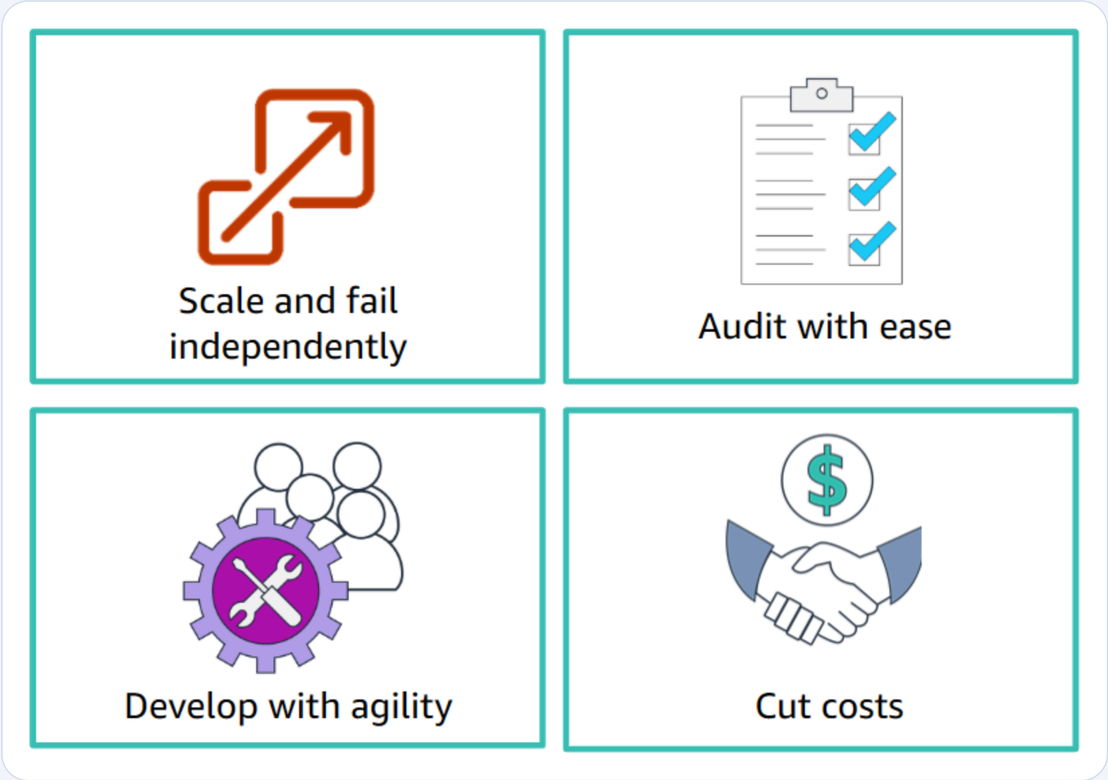
Event-driven architecture diagram: the order



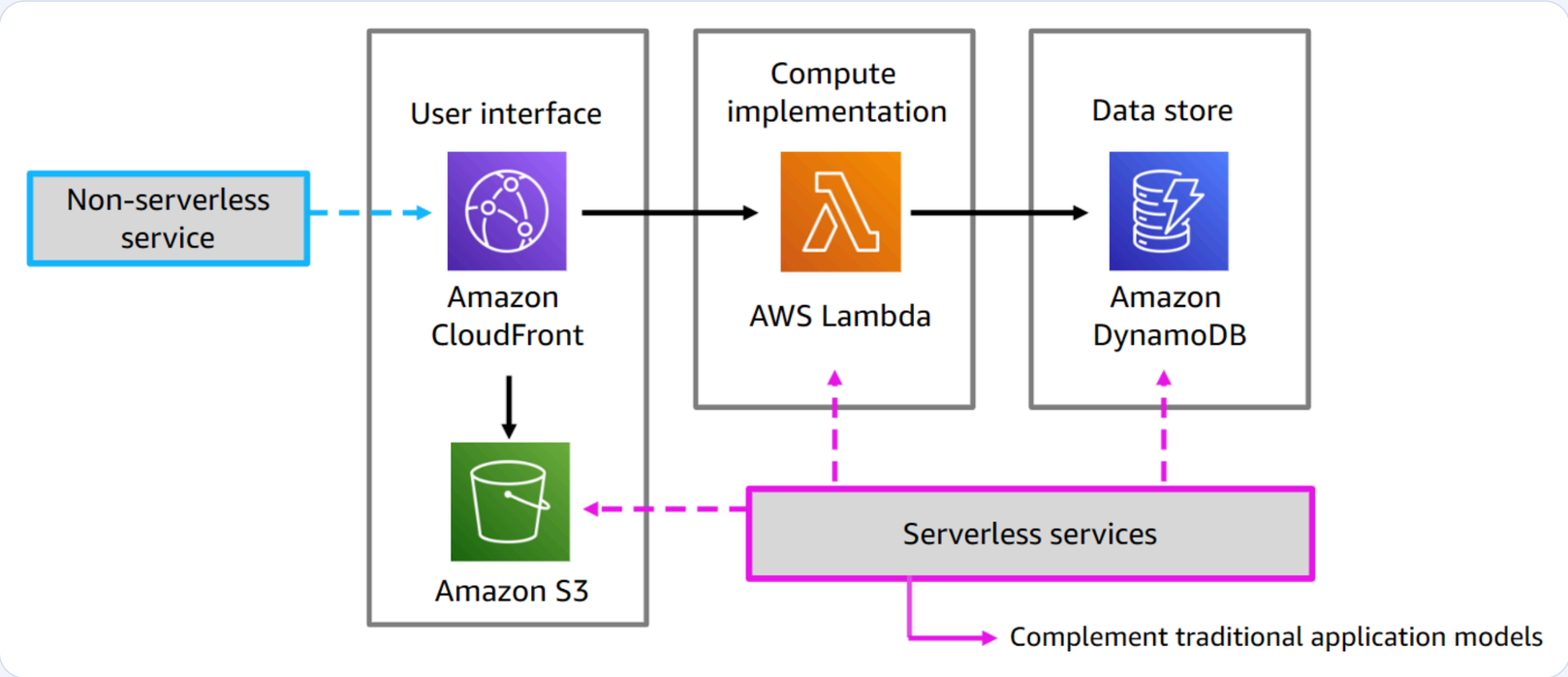
Event-driven architecture diagram: the flow



Event-driven architecture benefits



Serverless and non-serverless services together



Serverless technologies can complement traditional application models.

Key takeaways

01

Microservices

An architecture that is loosely coupled is referred to as a microservices architecture.

02

Serverless

With serverless computing, you can build and run applications and services without thinking about servers.

03

Event-driven

An event-driven architecture uses events to initiate actions and communication between decoupled services.

04

Hybrid

Serverless technologies can complement traditional application models.

Next session



01

Security and compliance